

DATA 3464: Fundamentals of Data Processing

Image Processing

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Topic overview

- Image representation
- File formats and compression
- Preprocessing for image recognition tasks

Resources used:

- [Computer Vision: Algorithms and Applications \(2nd edition\)](#)
- [Pillow Documentation](#)
- [A blog post on image compression](#)
- [JPEG demo](#)

Images as 2D signals

- The light that enters a camera can be modelled as continuous signal:

$$f(x, y), \quad -\infty < x, y < \infty$$

- Digital images are sampled:

$$f[n, m], \quad n = n\Delta_x, m = m\Delta_y$$

where typically $\Delta_x = \Delta_y$

- The area $\Delta_x \times \Delta_y$ is called a **picture element**, or **pixel**

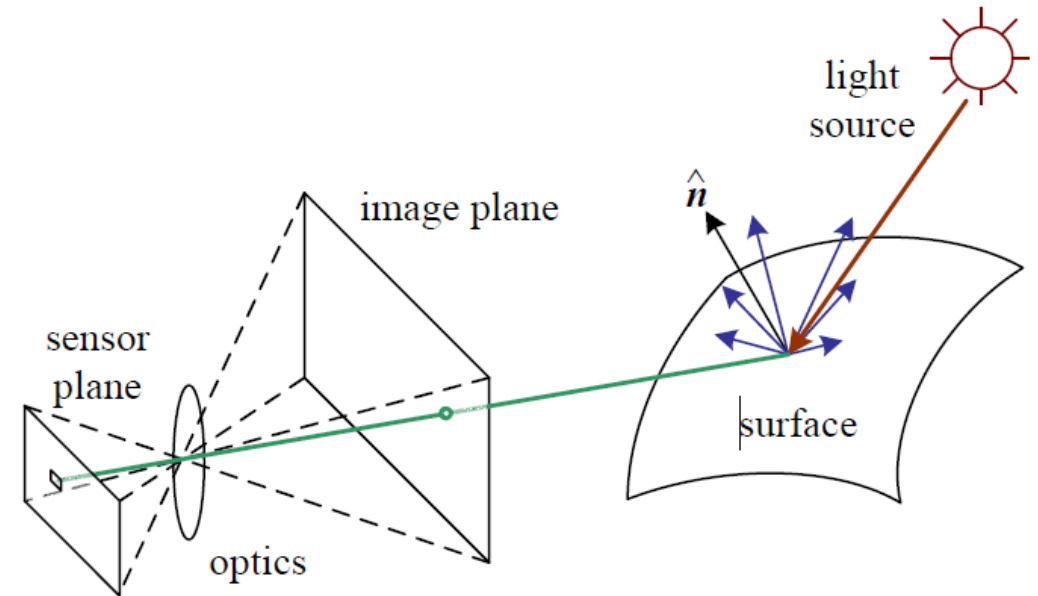
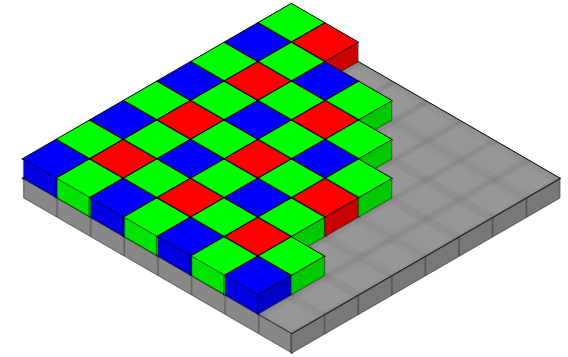


Image channels

- A photosensor responds to **light intensity** with an electrical signal
- Physical filters restrict the colour that reaches each sensor
- On most digital cameras, 1 sensor $\neq$ 1 pixel; it is instead **interpolated** to create a typical **3-channel image**
- ([Inexplicably](#), the Python imaging library Pillow uses the term "band")



Images as (3D) 2D arrays

- Size (`np.shape`) usually defined as `[rows, cols, (channels)]`
- Typical 8-bit image: integers from 0 to 255
 - For processing, common to convert to float and **normalize**
- RGB examples of individual pixels:
 - `[255, 0, 0]`
 - `[128, 128, 128]`
 - `[255, 255, 255]`

Portable Network Graphics

- PNG is an image format released in 1996 as a GIF replacement
- Typically 8 bits per channel, but can be 1, 2, 4, 8, or 16
- 3x RGB channels, plus an optional alpha (transparency) channel
- Space saved in two ways:
 - Indexed colour, where numbers map to a finite set of colours
 - Lossless compression, e.g. `[0 0 0 0 0 0 0]` → `7 0`

What kinds of images would benefit from this compression scheme?

Joint Photographic Experts Group

- **JPEG** is an image standard released in 1992 for photographic images
- **Lossy** compression based on human vision:
 - i. Channels converted to **YCbCr** colourspace
 - ii. The colour components (Cb/Cr) are downsampled 2-3x
 - iii. Blocks of 8x8 pixels are transformed to frequency domain
 - iv. Amplitudes are **quantized** with larger buckets at higher frequencies
 - v. Finally, lossless compression is applied
- The "quality" of a JPEG relates the degree of quantization

Where we left off on March 24

A Cautionary Tale of Image Classification

- As convolutional neural networks (CNNs) gained popularity as image recognition powerhouses, people started using them for all sorts of things
- One example: predicting whether someone is a criminal based on a photo
- **Thoroughly debunked** in a paper that mentions, among other issues:
 - Criminal samples are all 8-bit greyscale PNG photographs of printed images, taken with the same camera
 - Non-criminals are RGB JPEGs from 5 different sources, converted to greyscale to be "compatible with mugshots from the criminal category"

What's the problem here?

Preparing images for machine learning

- As with audio, images should be **consistent**
- Let's check out some [public datasets](#)
- Two approaches:
 - keep raw data and process on the fly (e.g. ImageNet)
 - apply some preprocessing and store (e.g. CIFAR-10)
- Common to do a bit of both, e.g. resize and save with consistent format, then apply various transforms during both training and inference

As usual, there is a tradeoff between time and space

Image Transformations

- **Geometric transforms** resize, rotate, skew, shift, etc
- **Point operators** independently change each pixel, e.g. histogram equalization
- **Linear filters** compute new pixel values from a weighted sum of values in a small neighbourhood
- **Nonlinear filters** compute new pixel values from a small neighbourhood in a nonlinear fashion

Resizing

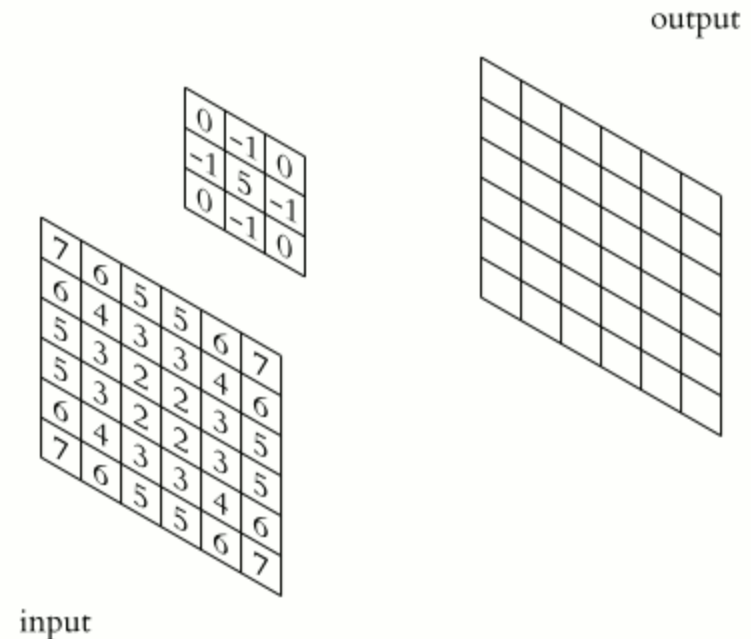
- For computer vision purposes, most images need to be resized
- Most often, a **square aspect ratio** is used
 - Can be rotated 90 degrees without modifying code
 - Simplifies parameter specification
 - Maybe some CUDA advantage if divisible by 8 (e.g. 224×224)
- Resizing needs **interpolation** (or resampling), and distorts aspect ratio
- Alternatively (or additionally), image **crops** can be used

Linear Filters: Convolution

- A filter or convolution **kernel** is a small (usually) square matrix that gets **convolved** with the image:

$$(f * g)[n, m] = \sum_{i=0}^n \sum_{j=0}^m f[i, j] g[n - i, m - j]$$

- Fun fact: this is the same as multiplication in frequency



Beyond linear filters

- Linear filters are simple weighted summations, and form the core of **convolutional neural networks**, widely used in computer vision
- Sometimes, nonlinear effects are useful, such as:
 - Median image filtering to reduce certain types of noise
 - Morphological operators for binary image masks

There is a huge world of image processing, including techniques for segmentation, feature detection, registration... too much to cover in this course!

Tools

In addition to Pillow and the usual numpy, matplotlib, scipy libraries:

- [ImageMagick](#) is a command-line tool that can be combined with bash scripting to process a bunch of images, e.g.:

```
mkdir resized
for img in *.jpg; do
    magick $img -resize 512x512! resized/$img
done
```

- [OpenCV](#) has functionality overlap with Pillow, but also computer vision algorithms
- [Torchvision](#) provides load-time transformations for use with deep learning models
- [ITK](#) is the OG classical image processing toolbox focused on medical imaging